COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/10/2026	**DATE SUBMITTED:**	07/15/2026
NAME:	Kherwin Emmanuel D. Milan	**INSTRUCTOR:**	Jimlord M. Quejado

Hands-on Activity 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

What procedures did you perform?

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	
Class Declaration and Method Definition Section	`class Triangle`
Main Function	`Triangle set1(40, 30, 110);`
Method Definition	`setAngles()`, and `validateTriangle()`

Table 1-1. C++ Structure Code for Answer

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a C++ program to swap the two numbers in different variables.

```
#include <iostream>

int main() {
    int firstNumber, secondNumber, temporary;

    std::cout << "Enter the first number: ";
    std::cin >> firstNumber;

    std::cout << "Enter the second number: ";
    std::cin >> secondNumber;

    std::cout << "Before swapping:\n";
    std::cout << "First number: " << firstNumber << "\n";
    std::cout << "Second number: " << secondNumber << "\n";
```

```
    temporary = firstNumber;
    firstNumber = secondNumber;
    secondNumber = temporary;

    std::cout << "After swapping:";
    std::cout << "First number: " << firstNumber << "\n";
    std::cout << "Second number: " << secondNumber << "\n";

    return 0;
}
```

```
Enter the first number: 4
Enter the second number: 3
Before swapping:
First number: 4
Second number: 3
After swapping:First number: 3
Second number: 4
PS D:\Downloads\Lab-20260714T121321Z-1-001>
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```
#include <iostream>

double convertToFahrenheit(double kelvin);

int main() {
    double kelvin;
    double fahrenheit;

    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> kelvin;

    if (kelvin < 0) {
        std::cout << "Kelvin cannot be negative." << std::endl;
    } else {
        fahrenheit = convertToFahrenheit(kelvin);

        std::cout << "Temperature in Fahrenheit: "
```

```
                  << fahrenheit << std::endl;
    }

    return 0;
}

double convertToFahrenheit(double kelvin) {
    return (kelvin - 273.15) * 9.0 / 5.0 + 32.0;
}
```

3. Create a C++ program that has a function that will calculate the distance between two points.

```
#include <iostream>
#include <cmath>

double calculateDistance(double x1, double y1,
                         double x2, double y2);

int main() {
    double x1, y1;
    double x2, y2;
    double distance;

    std::cout << "Enter x1: ";
    std::cin >> x1;

    std::cout << "Enter y1: ";
    std::cin >> y1;

    std::cout << "Enter x2: ";
    std::cin >> x2;

    std::cout << "Enter y2: ";
    std::cin >> y2;

    distance = calculateDistance(x1, y1, x2, y2);
```

```
    std::cout << "Distance between the two points: "
              << distance << std::endl;

    return 0;
}

double calculateDistance(double x1, double y1,
                         double x2, double y2) {
    double distance;

    distance = std::sqrt(
        ((x2 - x1) * (x2 - x1)) +
        ((y2 - y1) * (y2 - y1))
    );

    return distance;
}
```

4. Modify the code given in ILO B and add the following functions
A. A function to compute for the area of a triangle
B. A function to compute for the perimeter of a triangle
C. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

Conclusion
In this activity, I recalled how to use variables, functions, function prototypes, conditions, formulas, and classes in C++. I also started using std::cout and std::cin instead of using namespace std;. This activity helped me review the basic concepts of C++

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: *Grammarly*

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

COMPUTER ENGINEERING

QUEZON CITY

Purpose of Use: **Grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used grammarly only to correct grammar, spelling, and punctuation. It did not replace my original thinking, analysis, authorship, or completion of the required work**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.